

Dual-Input Priority-Switched Buck-Boost Power Adapter — Design Spec

Application: Compact power adapter for vintage film camera equipment. Low-duty-cycle load — short bursts, not continuous high-current operation. Board lives inside a small enclosure, so thermal path matters more than raw cooling capacity.

1. Physical

Parameter	Spec
Max PCB footprint	45mm × 75mm (shrunk from 85×45mm; length reduced, width unchanged)
Input connectors	M3-threaded, via solder-on SMT threaded standoff × 2 (Input 1, Input 2), for brass ring terminals
Output connectors	M3-threaded, via solder-on SMT threaded standoff × 2 (V+, GND), for brass ring terminals
Mounting holes	4× 3mm diameter, on a 70mm × 40mm rectangular pattern (recalculated from 80×40mm to fit the new footprint, same 2.5mm edge margin)
Thermal path (bottom)	Exposed metal pad on the bottom of the board, 45mm × 40mm, shifted as far toward the input side as possible while keeping a 7mm clearance from the center of the input-side mounting holes, thermally vias down from the LT8390 exposed pad and output MOSFETs, for direct contact with the enclosure floor as a heat sink
Thermal path (top)	Aluminum finned heatsinks on top of hot components wherever clearance allows —

Parameter	Spec
	LT8390, output MOSFETs, LTC4417's switching MOSFETs
Max total assembled height	Under 13.5mm, ideally — can flex up to 18mm if needed (PCB + tallest component/heatsink stack)
Indicators	Green Power LED, Red Fault LED — both on the bottom side of the board, at the output side

Flag for EE — threaded connector holes: Use solder-on SMT threaded standoffs (e.g. Keystone's SMT threaded insert line, or equivalent) rather than a heat-set insert — these solder onto the PCB pad during normal reflow/assembly, which fits JLCPCB's standard SMT process instead of requiring a separate manual heat-set step after the board is populated. Typical footprint: ~4.2-4.25mm mounting hole, ~6.2mm solder pad diameter, M3 female thread. Note the M3 designation describes the thread the standoff accepts, not the PCB hole size — the hole itself needs to be sized for the standoff body, not literally 3mm. "Zero height" variants exist that sit nearly flush with the board, worth considering given the height budget. Four total (2 input, 2 output).

Flag for EE: 45×75mm is noticeably tighter than the previous 85×45mm footprint for the same component set — the 6-8A-rated buck-boost inductor, LT8390 QFN, four switching MOSFETs, LTC4417 (or equivalent) plus its two P-channel MOSFETs, input fuses, the output voltage-selection jumper, input/output capacitors, two indicator LEDs, and the 70mm mounting-hole span all now need to fit in 10mm less length. Worth doing a rough component placement pass early — this is a real risk area, not just a formality, given the space that was freed up by moving to jumper-selected presets is now being given back to shrink the board.

Flag for EE — height budget: 13.5mm total is tight once you stack PCB thickness, component height, and a finned heatsink on top of that. The power inductor itself is often one of the tallest parts on a board like this (commonly several mm), and adding aluminum fins on top of the LT8390/MOSFETs adds more on top of whatever headroom is left. Worth sourcing low-profile finned heatsinks specifically (rather than generic tall ones) and doing a stack-up check — PCB + component + heatsink — early, since this height target and the bottom-side thermal pad both compete for the same vertical space. 18mm is the fallback ceiling if 13.5mm can't be hit cleanly, so there's some breathing room before this becomes a real problem.

2. Electrical — Input Stage

Parameter	Spec
Input 1 voltage range	11V–36V
Input 2 voltage range	11V–36V
Input priority	Input 2 preferred whenever valid; automatic fallback to Input 1 when Input 2 is absent or invalid. Not highest-voltage-wins — this must be a fixed priority, not a passive diode-OR.
Undervoltage cutoff	11V per input channel
Reverse polarity protection	Required on both inputs independently
Overcurrent protection	Fuse on each input line
Input summing	Not required — only one input powers the board at a time

Recommended part: LTC4417 (or equivalent prioritized PowerPath controller). It selects a supply based on pin assignment rather than voltage magnitude — wire Input 2 into the highest-priority pin and Input 1 into the next, and it will always prefer Input 2 when present. It operates over 2.5V–36V, includes reverse-battery protection to –42V, and switches via external back-to-back P-channel MOSFETs, giving reverse polarity protection on both inputs without extra discrete protection circuitry. The 11V UVLO should be set as the undervoltage threshold on each input channel (adjustable via a single external resistor per the datasheet) so an out-of-spec source is disqualified before it ever reaches the buck-boost stage.

Fuse placement: One fuse in series on each input line, ahead of the LTC4417. Rating should be sized off the final worst-case input current (output power ÷ minimum input voltage, with efficiency and margin factored in) once component selection is finalized — flagged as an open item below. A resettable PTC fuse is worth considering over a one-time blow fuse given this is field-used camera gear, but either is workable.

3. Electrical — Output Stage

Parameter	Spec
Output voltage	Preset-selectable via short/jumper terminal: 12V or 24V (discrete steps, not continuously variable)
Output current (design rating)	6–8A
Real-world duty cycle	~2A continuous for up to 10 min, or ~4A bursts for up to 3 min max

Recommended part: LT8390 synchronous 4-switch buck-boost controller, paired with 4 external N-channel MOSFETs. It covers a 4V–60V input and 1V–60V output range and is rated to deliver well over 100W with this topology, comfortably covering the target power range with margin.

Voltage selection: Implement via short/jumper-selectable taps on the LT8390's FB feedback resistor divider — each jumper position (or short-circuit terminal setting) corresponds to a fixed preset output voltage, rather than a continuously variable trim pot. This trades continuous adjustability for a smaller footprint, lower height, and a setting that can't drift or get bumped once selected — a real plus for gear that lives inside a sealed enclosure in the field. Presets: 12V and 24V. Note for EE: confirm the compensation network stays stable at both preset voltages, since buck-boost loop dynamics can shift with output voltage.

4. Protection & Indicators

Feature	Approach
Power LED	Green. Lit whenever regulated output is present. Placed on the bottom side of the board, at the output side
Fault LED	Red. Driven from the LT8390's PGOOD pin — it's an open-drain output pulled low whenever the output falls outside $\pm 10\%$ of the regulation target, so it can sink an LED directly through a pull-up resistor. Placed on

Feature	Approach
	the bottom side of the board, at the output side
IC-level over-temperature protection	Built into the LT8390: internal thermal shutdown halts switching if the die exceeds its threshold, and automatically resumes once it cools
Board-level over-temperature protection	The IC's internal thermal shutdown protects the controller itself, not the external power MOSFETs or inductor, which are the bigger heat sources in a 4-switch topology. Recommend adding an NTC thermistor (or thermal cutoff switch) mounted near the MOSFETs/inductor, wired to pull the LT8390's EN/UVLO pin low above a set board temperature — this extends over-temp protection to the whole power stage, not just the IC. Latching: once tripped, stays shut down until the input power is physically cycled (unplugged and reconnected) — no auto-recovery. Fault LED should latch along with it, so the indicator stays lit through the outage rather than only flashing momentarily.

5. Thermal Design

The rated 6–8A output is a design ceiling, not the expected operating point — actual use is short bursts at 2–4A, which gives real thermal margin versus a continuous-duty design at the same rating. That said, since the board sits in a small sealed enclosure:

- Thermal vias under the LT8390's exposed pad, routed to a bottom-layer copper pour
- Same treatment for the output-stage power MOSFETs
- Bottom-side pour terminates in an exposed metal pad for direct contact with the enclosure floor as the primary heat sink
- Low-profile aluminum finned heatsinks on top of the LT8390, output MOSFETs, and LTC4417 switching MOSFETs wherever component spacing and the 13.5mm height budget allow — dual-sided cooling (bottom conduction to the enclosure, top convection via fins) rather than relying on one path alone

- Board-level NTC/thermal cutoff (Section 4) monitors the hottest point near the inductor/MOSFETs as a backstop beyond passive conduction cooling
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6. Open Items for EE to Confirm

- Fuse current rating, once final component selection and worst-case input current are known
 - Fuse type: one-time blow vs. resettable PTC
 - Enclosure thermal interface method for the bottom metal pad — thermal pad material, direct copper-to-metal contact, or something else
 - Indicator LED colors/behavior (e.g., solid vs. blinking fault indication)
 - Confirm loop stability of the feedback network at both 12V and 24V preset points
 - Confirm feasibility of the 6-8A inductor + full component set (including fuses, voltage-selection jumper, LEDs) within the tighter 45×75mm footprint before committing to layout — this is a meaningfully tighter fit than the previous 85×45mm version, worth an early placement check rather than assuming it'll work out
 - Confirm PCB + component + heatsink stack-up fits under the 13.5mm height target, and identify which component (likely the inductor or a heatsink) is the limiting factor
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7. Mechanical Drawing

See the accompanying [power-adapter-mechanical-drawing.svg](#) for the dimensioned top-view drawing: board outline, mounting hole pattern, input side / output side labeling, bottom-side thermal pad footprint and offset, and LED locations. Individual input/output connector hole positions aren't fixed in this drawing — it marks the side zones only, exact placement of the four threaded M3 holes within each side is the EE's call. This is a placement schematic, not a fabrication-ready drawing — final component layout, trace routing, and clearances are the EE's call.

8. Suggested Architecture Summary

Input 1 (11-36V) —[Fuse]—

└─→ [LTC4417 PowerPath: priority select, reverse protect] ─→ [LT8390
4-switch buck-boost] ─→ 12V/24V preset / 6-8A out

Input 2 (11-36V) —[Fuse]—┐ (Input 2 = highest priority pin) |
Power LED, Fault LED (PGOOD)
NTC over-temp cutoff → EN/UVLO

Two-IC front-end/back-end split, both Analog Devices/Linear Tech parts with well-documented reference designs — should be straightforward for an EE to pick up and lay out.